① What is Numerical Techniques?

-> Numerical technique is an approach for solving complex mathmatical problems using simple arithmatic operations.

-> It involves formulations of mathmatical model, algorithms and computation techniques to approximate solutions to mathmatical problems.

② What is floating point number?

> A floating point number is a type of number used in computer programming that is used to represent real numbers with a high degree of precision.

55.66 can be represented as 5.566×10^1

0.5566×10^2

0.05566×10^3 <- Exponent

Significant (Mantissa)

Base

③ Floating point conversion:

-> In numerical technique, conversion is the process of changing a number from one base to another.

For example- Convert a binary number to decimal or decimal to binary, octal and hexa-decimal.

$(111.011)_2 = (7.375)_{10}$

$111 = 2^0 \times 1 + 2^1 \times 1 + 2^2 \times 1 = 1 + 2 + 4 = 7$

$011 = 0 \times \frac{1}{2^1} + 1 \times \frac{1}{2^2} + 1 \times \frac{1}{2^3}$

$= 0 + \frac{1}{4} + \frac{1}{8}$

$= \frac{3}{8} = 0.375$

Q. $(10100.101)_2 = (20.625)_{10}$

$10100 = 2^0\times0 + 2^1\times0 + 2^2\times1 + 2^3\times0 + 2^4\times1$

$= 4 + 16$

$= 20$

$101 = 1\times\frac{1}{2^1} + 0\times\frac{1}{2^2} + 1\times\frac{1}{2^3}$

$= \frac{1}{2} + \frac{1}{8}$

$= \frac{5}{8} = 0.625$

Q. $(101.101)_2 = (5.625)_{10}$

$101 = 2^0\times1 + 2^1\times0 + 2^2\times1$

$= 1 + 4$

$= 5$

$101 = 1\times\frac{1}{2^1} + 0\times\frac{1}{2^2} + 1\times\frac{1}{2^3}$

$= \frac{1}{2} + \frac{1}{8} = \frac{5}{8} = 0.625$

Q. $(101110.0011)_2 =$

$101110 = 2^0\times0 + 2^1\times1 + 2^2\times1 + 2^3\times1 + 2^4\times0 + 2^5\times1$

$= 0 + 2 + 4 + 8 + 0 + 32$

$= 45$

$0011 = 0\times\frac{1}{2^1} + 0\times\frac{1}{2^2} + 1\times\frac{1}{2^3} + 1\times\frac{1}{2^4}$

$= \frac{1}{8} + \frac{1}{16}$

$= \frac{2+1}{16}$

$= \frac{3}{16}$

$=$

Q. $(0.859375)_{10} = (0.110111)_2$

$0 = 2^{-1} \times 0 = 0 \;/\; 0 \times \frac{1}{2^1} = 0$

```
   .859375
    X    2
  ---------
1  .718750
    X    2
  ---------
1  .437500
    X    2
  ---------
0  .875,000
    X    2
  ---------
1  .750000
```

```
   .750000
    X    2
  ---------
1  .500000
    X    2
  ---------
1  .000000
```

Q. $(0.7)_{10} = (0.1011001100)_2$

Find its round off error.

-> Round-off error is the difference between the exact result of a calculation and the result of the same calculation using rounded numbers.

$0 = 2^{-1} \times 0 = 0 \;/\; 0 \times \frac{1}{2^1} = 0$

```
   .7
   X2
  ---
1  .4
   X2
  ---
0  .8
   X2
  ---
1  .6
   X2
  ---
1  .2
   X2
  ---
0  .4
   X2
  ---
0  .8
```

```
   .6
   X2
  ---
1  .2
   X2
  ---
0  .4
   X2
  ---
0  .8
```

$(0.\overline{1011001100})_2$

$\Rightarrow 1\times\frac{1}{2^1}+0\times\frac{1}{2^2}+1\times\frac{1}{2^3}+1\times\frac{1}{2^4}+0\times\frac{1}{2^5}+0\times\frac{1}{2^6}+1\times\frac{1}{2^7}$

$\Rightarrow \frac{1}{2}+\frac{1}{8}+\frac{1}{16}+\frac{1}{128}$

$\Rightarrow \frac{64+16+8+1}{128}$

$\Rightarrow 0.6953125$

Round-off error $= (0.7)-(0.6953125)$

$= 0.0046875$

Q. $(28.125)_{10} = (11100.001)_2$

2	28	
2	14	0
2	7	0
2	3	1
2	1	1
	0	1

	.125
	X 2
0	.250
	X 2
0	.500
	X 2
1	.000
	X 2
0	.000

Q. $(43.167)_{10} = (101011.00101)_2$

2	43	
2	21	1
2	10	1
2	5	0
2	2	1
2	1	0
	0	1

	.167
	X 2
0	.334
	X 2
0	.668
	X 2
1	.336
	X 2
0	.672

	.672
	X 2
1	.344
	X 2
0	.688
	X 2
1	.376
	X 2
0	.752

Q. Convert the binary number $(1101001.1110011)_2$ into octal and hexa-decimal.

$(\overleftarrow{1101001}.\overrightarrow{1110011})_2 = (105.898)_{10}$

$1101001 = 2^0\times1 + 2^1\times0 + 2^2\times0 + 2^3\times1 + 2^4\times0 + 2^5\times1 + 2^6\times1$

$\Rightarrow 2 + 8 + 32 + 64$

$\Rightarrow 105$

$1110011 = 1\times\frac{1}{2^1} + 1\times\frac{1}{2^2} + 1\times\frac{1}{2^3} + 0\times\frac{1}{2^4} + 0\times\frac{1}{2^5} + 1\times\frac{1}{2^6} + 1\times\frac{1}{2^7}$

$= \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{64} + \frac{1}{128}$

$= \frac{64+32+16+2+1}{128}$

$= \frac{115}{128} = 0.898$

$(105.898)_{10} = (151.713)_8$

8	105	
8	13	1
8	1	5
	0	1

$.898 \times 8 = 7.184$

$.184 \times 8 = 1.472$

$.472 \times 8 = 3.776$

$(105.898)_{10} = (69.E6)_{16}$

16	105	
16	6	9
	0	6

$.898 \times 16 = 14.368$

$.368 \times 16 = 5.888$

Q. $(658.825)_{10} = (1222.6463)_8$

8	658	
8	82	2
8	10	2
8	1	2
	0	1

$.825 \times 8 = 6.600$

$.600 \times 8 = 4.800$

$.800 \times 8 = 6.400$

$.400 \times 8 = 3.200$

$(658.825)_{10} = (292.D3)_{16}$

16	658	
16	41	2
16	2	9
	0	2

$.825 \times 16 = 13.200$

$.200 \times 16 = 3.200$

Q. $(239.563)_{10} = (357.440)_8$

8	239	
8	29	7
8	3	5
	0	3

$.563 \times 8 = 4.504$

$.504 \times 8 = 4.032$

$.032 \times 8 = 0.256$

$(239.563)_{10} = (1415.902)_{16}$

$(EF.902)_{16}$

16	239	
16	14	15
	0	14

$.563 \times 16 = 9.008$

$.008 \times 16 = 0.128$

$.128 \times 16 = 2.048$

Error:

In numerical techniques, error is the difference between the true value and the approximate value obtained.

There are several types of errors that can occur in numerical techniques are:

① Computational error
② Truncation error

① Computational error:

Computational errors are the errors that occur in the computational data, such as in vectors and matrices, during a computation. It does not produce the expected result.

i> Round-off error:

Rounding off means a number is made simpler by keeping its value intact but closer to the next numbers.

Rounding rules for decimal numbers are follows:

-> Determine the rounding digit and look at its right hand side.
-> If the digit on the right-hand side is less than 5, consider them as equal to zero / as it.
-> If the digit on the right hand side is greater than 5, then add +1 to that digit.
-> If the digit on the right-hand side is equal to 5, then we see previous digit is even or odd.
-> If the previous digit is even then the digit remains same or if the previous digit is odd then add +1 to that digit.

Eg- Round-off up to 4 decimal places

0.567246 = 0.5672
0.592788 = 0.5928

<5 = same
>5 = +1

0.363553 = 0.3636 =5 previous odd = +1

0.432256 = 0.4322 =5 previous even = same

ii) Significant error:

There are three main sources of errors in numerical Computation: Rounding, data uncertainty and truncation. We find in decimal number how many significant digit in this number. we count it by starting number without zero i.e starting zero is not countable.

Ex-

How many significant digits are these:

1.2023 = 5 significant digit

0.1123 = 4 " " "

1.000 = 4 " " "

0.00001 = 1 significant digit

iii) Scientific error form:

Scientific notation is a method for expressing a given quantity as a number having significant digits necessary, multiplied by 10 to the appropriate power such as 1.56×10^7. It is a form of presenting very large numbers or very small numbers in a simpler form.

$0.0367895 \times 10^5 = 3.67895 \times 10^3$

$0.000000084 = 8.4 \times 10^{-8}$

* Absolute error

$$\varepsilon_A = |x_T - x_A|$$

* Relative error

$$\varepsilon_R = \left|\frac{x_T - x_A}{x_T}\right| = \left|1 - \frac{x_A}{x_T}\right|$$

* Percentage error $\varepsilon_p = \varepsilon_R \times 100$

② Truncation Error:

Truncation error is the error defined as the difference between the true derivative of a function and its derivative obtained by numerical aproximation.

There are two types of truncation error;

i) Local Truncation error
ii) Global Truncation error

i) Local Truncation error:

- Local Truncation error is defined as the error introduced in performing one step of the numerical method.
- The error caused by one iteration.
- Local truncation error represents the differential equation and its FD representation at a point in space and time.

ii) Global Truncation error:

- Global truncation error is defined as the error introduced in performing many step of the numerical method.
- The global error caused by more than one iterations.
- Global truncation error refers to the error that affect overall sentence organization / overall iterations.
- Global truncation error is the total error that occurs when an approximate numerical solutions differs from the actual solution over the entire domain of a problem.

Order of a method:

→ The order of a numerical method is a measure of how the error of a numerical solution changes as the step size decreases.

order 1 - Bisection method, Regula-falsi

order 2 - Newton Raphson method, Newtons method for solving non-linear system.

Convergence and terminal conditions:

Convergence:- In numerical method convergence is a property that ensure the sequence of value produced by a numerical method gets closer to the true solution as the number of iteration increases.

→ Convergence is a condition on the numerical solution that ensures the output of the simulation is a correct representation of the model / gutient / function.

→ A final difference aproximation is considered as consistent if the truncation error terms approaches zero when the grid or mesh and time state size are reduced.

→ Sequences with larger orders converges more quickly than those with smaller order.

→ A numerical method is covergent if and only if a sequence of a method solutions with increasingly refined solution domains approaches a fined value.

CM- The serieses $\sum_{n=1} \frac{1}{2^n} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} \cdots \cdots \cdots n$

It is convergent because the partial sum get closer and closer to one.

Bisection method:

→ The Bisection method is a numerical root finding technique used to solve equations that lack analytical solutions and to find the roots of polynomial equations. It works by dividing an interval into smaller intervals and narrowing down the range of possible solutions iteratively.

→ The bisection method can be used to solve the equation $f(x)=0$ for a real variable x, given that f is a continous function defined on the interval $[a, b]$ and $f(a)$ and $f(b)$ have opposite signs.

→ Calculate the midpoint, t, between a and b.

$$t = \frac{a+b}{2}$$

→ If $f(t)$ equals 0, then t is the root of the function. If not proceed to the next step.

→ Divide the interval $[a,b]$:

- $f(a)$ and $f(t)$ exhibit different signs and enclose a root within their range.
- $f(c)$ and $f(b)$ display different signs and enclose a root within their range.

→ A sub-interval is selected as the new interval for the subsequent step, and this repetition continues until the interval reaches a suitable level of smallness.

→ Continue this procedure until we obtain a sequence of nested steps of sub-interval $I \supset I_1 \supset I_2 \cdots \cdots \cdots$ such that each sub interval contains the root.

Q. Find the real root of the equation $x^3 - x - 11 = 0$ by bisection method, correct to three places of decimal.

Let $f(x) = x^3 - x - 11$

$f(0) = 0^3 - 0 - 11 = -11$ (-ve)

$f(1) = 1^3 - 1 - 11 = -11$ (-ve)

$f(2) = 2^3 - 2 - 11 = -5$ (-ve)

$f(3) = 3^3 - 3 - 11 = 13$ (+ve)

The smallest possitive root lies in the interval (2,3)

First approximation to root

$$m_1 = \frac{2+3}{2} = \frac{5}{2} = 2.5$$

$f(m_1) = (2.5)^3 - 2.5 - 11$

$= 15.625 - 2.5 - 11$

$= 2.125$

Thus the root lies in the interval taking $a_1 = 2$, $b_1 = 2.5$ are of opposite signs.

2nd approximation to root

$$m_2 = \frac{2+2.5}{2} = \frac{4.5}{2} = 2.25$$

$f(m_2) = (2.25)^3 - 2.25 - 11$

$= 11.3906 - 2.25 - 11$

$= -1.8594$

Thus the root lies in the interval $a_2 = 2.25$, $b_2 = 2.5$ are of opposite signs.

3rd approximation to root

$$m_3 = \frac{2.25 + 2.5}{2} = 2.375$$

$f(m_3) = (2.375)^3 - 2.375 - 11$

$= 13.3964 - 2.375 - 11$

$= 0.0214$

Thus the root lies in the interval $a_3 = 2.25$, $b_3 = 2.375$ are of opposite signs.

4th approximation to root

$$m_4 = \frac{2.25 + 2.375}{2} = 2.3125$$

$$f(m_4) = (2.3125)^3 - 2.3125 - 11$$
$$= 12.3664 - 2.3125 - 11$$
$$= -0.9461$$

Thus the root lies in the interval $a_4 = 2.3125, b_4 = 2.375$ are of opposite signs.

5th approximation to root

$$m_5 = \frac{2.3125 + 2.375}{2} = 2.34375$$

$$f(m_5) = (2.34375)^3 - 2.34375 - 11$$
$$= 12.8746 - 2.34375 - 11$$
$$= -0.46915$$

Thus the root lies in the interval $a_5 = 2.34375, b_5 = 2.375$ are of opposite signs.

6th approximation to root

$$m_6 = \frac{2.34375 + 2.375}{2} = 2.3593$$

$$f(m_6) = (2.3593)^3 - 2.3593 - 11$$
$$= 13.1325 - 2.3593 - 11$$
$$= -0.2268$$

Thus the root lies in the interval $a_6 = 2.3593, b_6 = 2.375$ are of opposite signs.

7th approximation to root

$$m_7 = \frac{2.3593 + 2.375}{2} = 2.36715$$

$$f(m_7) = (2.36715)^3 - 2.36715 - 11$$
$$= 13.26408 - 2.36715 - 11$$
$$= -0.10307$$

Thus the root lies in the interval $a_7 = 2.36715$, $b_7 = 2.375$ are of opposite signs.

8th approximation to root

$$m_8 = \frac{2.36715 + 2.375}{2} = 2.37107$$

$$f(m_8) = (2.37107)^3 - 2.37107 - 11$$
$$= 13.33009 - 2.37107 - 11$$
$$= -0.04098$$

Thus the root lies in the interval $a_8 = 2.37107$ $b_8 = 2.375$ are of opposite signs.

9th approximation to root

$$m_9 = \frac{2.37107 + 2.375}{2} = 2.37303$$

$$f(m_9) = (2.37303)^3 - 2.37303 - 11$$
$$= 13.3171 - 2.37303 - 11$$
$$= -0.0532$$

Thus the root lies in the interval $a_9 = 2.37303$ $b_9 = 2.375$ are of opposite signs.

10th approximation to root

$$m_{10} = \frac{2.37303 + 2.375}{2} = 2.374015$$

$\therefore$ Root = 2.374

Q. Perform 5 iterations of the Bisection method to obtain the smallest positive root of the equation $f(x) = x^3 - 5x + 1 = 0$.

$f(x) = x^3 - 5x + 1$

$f(0) = 0 - 0 + 1 = 1$ (+ve)

$f(1) = 1^3 - 5(1) + 1 = -3$ (-ve)

Thus the smallest positive root lies in the interval $a = 0$, $b = 1$ are of opposite signs. 1st approximation to root -

$$m = \frac{0 + (1)}{2} = \frac{1}{2} = 0.5$$

$f(m) = (0.5)^3 - 5(0.5) + 1$

$= (0.125) - 2.5 + 1$

$= -1.375$

Thus the root lies in the interval $a_1 = .5$ $b_1 = 1$ are of opposite signs.

2nd approximation to root -

$$m_1 = \frac{0.5 + 0}{2} = 0.25$$

$f(m_1) = (0.25)^3 - 5(0.25) + 1$

$= 0.0156 - 1.25 + 1$

$= -0.2344$

Thus the root lies in the interval $a_2 = 0.25$, $b_2 = 0$ are of opposite signs.

3rd approximation to root

$$m_2 = \frac{0.25 + 0}{2} = 0.125$$

$f(m_2) = (0.125)^3 - 5(0.125) + 1$

$= 0.001953 - 0.625 + 1$

$= 0.37695$

Thus the root lies in the interval $a_3 = 0.25$, $b_3 = 0.125$ are of opposite signs.

4th approximation to root

$$m_3 = \frac{0.25+0.125}{2} = \frac{0.375}{2} = 0.1875$$

$$f(m_3) = (0.1875)^3 - 5(0.1875) + 1$$
$$= 0.00659 - 0.9375 + 1$$
$$= 0.069$$

Thus the root lies in the interval $a_4 = 0.25$, $b_4 = 0.1875$ are of opposite signs..

5th approximation to root

$$m_4 = \frac{0.25+0.1875}{2} = \frac{0.4375}{2} = 0.2187$$

$$f(m_4) = (0.2187)^3 - 5(0.2187) + 1$$
$$= 0.01046 - 1.0935 + 1$$
$$= -0.08304$$

K	a_{k-1}	b_{k-1}	mp	$f(m_k) \times (f(m_{-1})$
1	0	1	0.5	<0
2	0	0.5	0.25	<0
3	0	0.25	0.125	>0
4	0.125	0.25	0.1875	>0
5	0.1875	0.25	0.21875	<0

Q. perform 5 iterations of the Bisection method to obtain a root of the equation of $f(x) = \cos x - xe^x = 0$.

$f(x) = \cos x - x e^x$

$f(0) = \cos 0 - 0 \times e^0$

$= 1 - 0 \times 1$

$= 1$ (+ve)

$f(1) = \cos 1 - 1 \times e^1$

$= 0.9998 - 1 \times 2.71828$

$= 0.9998 - 2.71828$

$= -1.71848$ (-ve)

Thus the smallest possitive root lies in the interval $a=0, b=1$ are of opposite signs.

1st approximation to root-

$$m = \frac{0+1}{2} = \frac{1}{2} = 0.5$$

$f(m) = \cos(0.5) - 0.5 \times e^{0.5}$

$= 0.9999 - 0.8243$

$= 0.1756 > 0$

Thus the root lies in the interval $a_1 = 1$ and $b_1 = 0.5$ are of opposite signs.

2nd approximation to root-

$$m_1 = \frac{1+0.5}{2} = \frac{1.5}{2} = 0.75$$

$f(m_1) = \cos(0.75) - 0.75 \times e^{0.75}$

$= 0.7316 - 0.75 \times 2.117$

$= -0.8561 < 0$

Thus the root lies in the interval $a_2 = 0.75$ and $b_2 = 0.5$ are of opposite signs.

3rd approximation to root-

$$m_2 = \frac{0.75+0.5}{2} = \frac{1.25}{2} = 0.625$$

$f(m_2) = \cos(0.625) - 0.625 \times e^{0.625}$

$= 0.8109 - 1.1676$

$= -0.3567 < 0$

Thus the root lies in the interval $a_3 = 0.625$ and $b_3 = 0.5$ are of opposite signs.

4th approximation to root-

$$m_3 = \frac{0.625+0.5}{2} = \frac{1.125}{2} = 0.5625$$

$$f(m_3) = \cos(0.5625) - 0.5625 \times e^{0.5625}$$
$$= 0.845924 - 0.5625 \times e^{0.5625}$$
$$= -0.1412 < 0$$

Thus the root lies in the interval $a_4 = 0.5625$ and $b_4 = 0.5$ are of opposite signs.

4th approximation to root-

$$m_4 = \frac{0.5625+0.5}{2} = \frac{1.0625}{2} = 0.53125$$

$$f(m_4) = \cos(0.53125) - 0.53125 \times e^{0.53125}$$
$$= 0.862174 - 0.53125 \times e^{0.53125}$$
$$= -0.0415 < 0$$

Sequence of intervals for the Bisection Method

k	a_{k-1}	b_{k-1}	m_k	$f(m_k) \times f(a_{k-1})$
1	0	1	0.5	>0
2	0.5	1	0.75	<0
3	0.5	0.75	0.625	<0
4	0.5	0.625	0.5625	<0
5	0.5	0.5625	0.53125	<0

Secant method :- / chord method :-

→ If x_{k-1} and x_k are two approximations to the root, then we determine a_0 and a_1 in $f(x) = a_0 x_{k-1} + a_1 = 0$ by using the conditions ($x = \frac{-a_1}{a_0}$)

$$f(x_{k-1}) = f(k-1) = a_0 x_{k-1} + a_1 \quad \text{——— (i)}$$

$$f(x_k) = f_k = a_0 x_k + a_1 \quad \text{——— (ii)}$$

On solving we obtain-

$$f_k - f_{k-1} = a_0 (x_k - x_{k-1})$$

$$\Rightarrow a_0 = \frac{f_k - f_{k-1}}{x_k - x_{k-1}}$$

$$x_{k+1} = \frac{x_{k-1} f_k - x_k f_{k-1}}{f_k - f_{k-1}}$$

Which may also be written as-

$$x_{k+1} = x_k - \frac{x_k - x_{k-1}}{f_k - f_{k-1}} f_k, \quad k = 1, 2 \cdots n$$

Q. A real root of the equation $f(x) = x^3 - 5x + 1 = 0$ lies in the interval $(0, 1)$, perform four iterations of the secant method.

$x_0 = 0$, $x_1 = 1$, $k = 1$

$$x_{k+1} = \frac{x_{k-1} f_k - x_k f_{k-1}}{f_k - f_{k-1}}$$

$$x_2 = \frac{x_0 f_1 - x_1 f_0}{f_1 - f_0}$$

$f(x) = x^3 - 5x + 1 = 0$

$f(0) = 1 > 0$

$f(1) = -3 < 0$

$$x_2 = \frac{0(-3) - 1(1)}{-3 - 1} = \frac{+1}{+4} = \frac{1}{4} = 0.25$$

$f(x_2) = (0.25)^3 - 5(0.25) + 1$

$= -0.234375$

$$x_3 = \frac{x_1 f_2 - x_2 f_1}{f_2 - f_1} = \frac{1(-0.234375) - 0.25(-3)}{-0.234375 - (-3)}$$

$$= \frac{0.515625}{2.765625} = 0.186441$$

$f(x_3) = (0.186441)^3 - 5(0.186441) + 1$

$= 0.074276$

$$x_4 = \frac{x_2 f_3 - x_3 f_2}{f_3 - f_2} = \frac{0.062266}{0.308651} = 0.201736$$

$f(x_4) = (0.201736)^3 - 5(0.201736) + 1$

$= -0.00047$

$$x_5 = \frac{x_3 f_4 - x_4 f_3}{f_4 - f_3} = \frac{-0.015072}{-0.074746} = 0.201654$$

Q. Use the secant method to perform five iterations of the equation $\cos x - x e^x = 0$.

Taking the initial approximation as $x_0 = 0$, $x_1 = 1$, $k = 1$

$f(0) = \cos 0 - 0 e^0 = 1 > 0$

$f(1) = \cos 1 - 1 e^1 = 0.540302 - 1 e^1 = -2.17797983 < 0$

$$x_{k+1} = \frac{x_{k-1} f_k - x_k f_{k-1}}{f_k - f_{k-1}}$$

$$x_2 = \frac{x_0 f_1 - x_1 f_0}{f_1 - f_0} = \frac{0(-2.17797983) - 1(1)}{-2.17797983 - 1}$$

$$= \frac{-1}{-3.17797983} = 0.3146653$$

$$f(x_2) = \cos(0.3146653) - 0.3146653 \times e^{0.3146653}$$
$$= 0.9509 - 0.3146653 \times e^{0.3146653}$$
$$= 0.519871$$

$$x_3 = \frac{x_1 f_2 - x_2 f_1}{f_2 - f_1} = \frac{1(0.519871) - 0.3146653(-2.17797983)}{0.519871 + 2.17797983}$$

$$= \frac{0.519871 + 0.685334677}{2.69785083} = 0.4467281$$

$$f(x_3) = \cos(0.4467281) - 0.4467281 \times e^{0.4467281}$$
$$= 0.901865 - 0.4467281 \times e^{0.4467281}$$
$$= 0.203544$$

$$x_4 = \frac{x_2 f_3 - x_3 f_2}{f_3 - f_2} = \frac{0.3146653(0.203544) - 0.4467281(0.51987}{0.203544 - 0.519871}$$

$$= \frac{0.064048 23 - 0.23224048}{-0.316327} = \frac{-0.16819275}{-0.316327}$$

$$= 0.531705$$

$$f(x_4) = \cos(0.531705) - 0.531705 \times e^{0.531705}$$
$$= 0.861944 - 0.531705 \times e^{0.531705}$$
$$= -0.0429283$$

$$x_5 = \frac{x_3 f_4 - x_4 f_3}{f_4 - f_3} = \frac{0.4467281(-0.0429283) - 0.531705(0.20354}{-0.0429283 - 0.203544}$$

$$= \frac{-0.01917728 - 0.10822536}{-0.2464723}$$

$$= \frac{-0.12740264}{-0.2464723} = 0.516904496$$